200ºC :

$$1\ atm \cdot \frac{473\ K}{293\ K} = 1{,}61\ atm$$

$2{,}88 - 1{,}61 = 1{,}27\ atm$.

, T :

$$P_{vap} = 1{,}27 \cdot \frac{T}{473}\ atm = \frac{T}{373}\ atm$$

$P(T)$.

:

$$P(T) = \frac{T}{373}\ atm,$$

.

$P(T)$ $P = 1\ atm, T = 373\ K = 100\ ℃$, .

채점 기준: 문제 1B:

№	평가 항목	배점
1.	: T=200℃	0,5
2.	T	0,5
3.	T_{vap}	0,5
4.		0,5
합계		**2.0**

문제 1.C 피라미드

1.

.

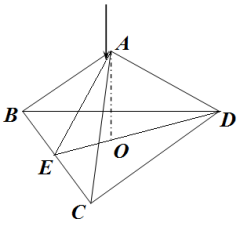
, $\angle AEO = \alpha$.

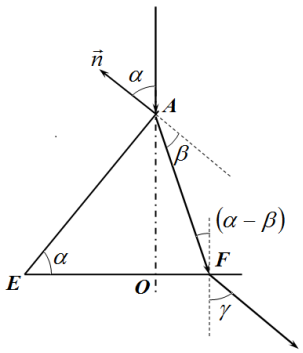
$$\cos\alpha = \frac{1}{\sqrt{3}}, \qquad \alpha = 54{,}7°.$$

β

:

$$\sin\beta = \frac{\sin\alpha}{n} = \frac{1}{n}\sqrt{\frac{2}{3}}. \qquad \beta = 24{,}6°.$$





F .

$$|AE| = \frac{a}{2},$$

$$|AO| = \frac{a}{2}\sin\alpha = \frac{a}{2}\sqrt{\frac{2}{3}} = 0{,}41mm.$$

$|OF| = |AO|tg(\alpha - \beta) = 0{,}24\ mm$.

:

$$|DF| = |OD| - |OF| = \frac{a}{\sqrt{3}} - 0{,}24mm = 0{,}91mm.$$

:

